Brian Wang

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Education

Worcester Polytechnic Institute **Aug 2024 – Dec 2027**
Bachelor of Science in Computer Science and Mathematical Sciences | GPA: 4.0 *Worcester, MA*

Relevant Coursework

- Algorithms
- Systems Programming
- Operating Systems
- Probability
- Stochastic Processes
- Linear Algebra
- Linear Programming
- Numerical Methods
- Artificial Intelligence
- Database Systems
- Computer Networks
- Compilers

Technical Skills

Languages: C++17/20, Python, Java, SQL, C
Systems / Tools: Linux, Git, AWS, REST APIs, QuantConnect
Data / ML: pandas, NumPy, scikit-learn, PyTorch, XGBoost

Experience

CYVL **May 2026 – Present**
Software Engineering Intern *Somerville, MA*

- Scoping R&D foundation for infrastructure deterioration prediction under sparse historical-data constraints.
- Analyzing public records, road-condition data, and LiDAR/video model outputs to evaluate scalable modeling approaches.

WPI Mathematical Sciences Department **Aug 2026 – May 2027**
Peer Learning Assistant *Worcester, MA*

- Selected by faculty to support undergraduate mathematics students; lead weekly problem-solving sessions and provide 1:1 instruction.

Projects

Single-Ticker Limit Order Book Matching Engine | *C++17, STL* **January 2026**

- Implemented single-symbol matching engine supporting limit orders, market orders, cancellations, partial fills, and FIFO price-time priority.
- Used ordered price levels and order-id lookup structures to support matching, best-price access, and cancellation paths.
- Benchmarked using randomized order events generated from Python workload scripts; measured per-order execution latency with RAII timers.
- Achieved 327K orders/sec throughput and 4.3µs P99 latency on M4 MacBook Air under reproducible synthetic load.

Algorithmic Trading Research and Live Deployment Study | *Python, QuantConnect* **May 2025 – Aug 2025**

- Developed 8 algorithmic trading strategies across momentum, mean reversion, statistical arbitrage, regime adaptation, and reinforcement learning.
- Deployed 4 strategies through QuantConnect paper trading over 5 weeks; analyzed live degradation against historical backtests.
- Evaluated robustness using Sharpe ratio, Probabilistic Sharpe Ratio, drawdown, and transaction-level performance analysis.
- Diagnosed live/backtest divergence caused by regime instability, event-driven volatility, and backtest-period selection.

Gamified Internship Tracker | *Python, SQL, AWS, REST* **Oct 2025 – Dec 2025**

- Led backend architecture for full-stack application in four-person Agile team across structured sprint cycles.
- Implemented REST APIs and authentication; iterated on reliability through stakeholder feedback.

Leadership / Extracurricular

WPI Swim Club – President (incoming) | WPI Student Investment Fund – Technology Sector Analyst | SASE – Peer Mentor